

OVERVIEW: CARBON DIOXIDE REMOVAL (CDR)

Carbon Dioxide Removal (CDR) refers to approaches that pull CO₂ directly from the atmosphere and store it durably in geological, terrestrial, or ocean reservoirs. Unlike traditional carbon capture—which stops emissions at the factory smokestack—CDR addresses historical emissions and cleans up ambient air.

1. BIOCHAR

Biochar is a charcoal-like substance created by heating biomass (such as agricultural waste, wood chips, or manure) in the absence of oxygen. This process is called pyrolysis.

How it Works: Plants naturally absorb CO₂ via photosynthesis. Pyrolysis locks that carbon into a stable, solid form. The resulting biochar is then buried or mixed into agricultural soils.

Storage Permanence: Hundreds to thousands of years, as the carbon is highly resistant to decomposition.

Co-benefits: Improves soil water retention, enhances nutrient availability for crops, and reduces the need for chemical fertilizers.

Limitations: Requires massive amounts of biomass feedstock. Improper production can release methane or other harmful pollutants.

2. DIRECT AIR CAPTURE (DAC)

DAC is an engineering-based approach that uses large industrial machines to extract CO₂ directly from ambient air.

How it Works: Giant fans pull air through a chemical filter. The filter uses liquid solvents or solid sorbents to bind with CO₂. The system is then heated to release the pure CO₂, which is compressed and injected deep underground into rock formations where it turns into stone.

Storage Permanence: Virtually permanent (tens of thousands of years) when combined with deep geological storage.

Co-benefits: Minimal land footprint compared to nature-based solutions. Facilities can be built on non-arable land near storage sites.

Limitations: Extremely energy-intensive. It requires significant low-carbon or renewable energy to run efficiently, and current costs per ton are very high.

3. ENHANCED ROCK WEATHERING (ERW)

ERW accelerates a natural geological process where rainwater absorbs CO₂ and reacts with rocks to dissolve them, trapping the carbon.

How it Works: Silicate rocks (like basalt) are finely crushed into dust to maximize their surface area. This dust is spread across large areas of agricultural land. When rain falls, the water reacts with the rock dust and atmospheric CO₂, turning the carbon into stable bicarbonate ions that eventually wash into the ocean.

Storage Permanence: Permanent (100,000+ years), as the carbon is locked away in ocean chemistry or sinks to the seafloor as carbonate minerals.

Co-benefits: Decreases soil acidity, releases vital nutrients (like magnesium and calcium) back into the soil, and helps combat ocean acidification.

Limitations: Highly dependent on mining, crushing, and transporting heavy rock, which consumes significant energy. Long-term impacts on local soil ecosystems are still being studied.

THE COST PER TON TO REMOVE CARBON DIOXIDE

Technology	Current 2026 Cost (per Metric Ton)	Target Cost by 2030–2050	Primary Cost Drivers
Biochar	\$100 – \$250	<\$100	Biomass logistics, pyrolysis hardware.
Enhanced Rock Weathering (ERW)	\$200 – \$500	\$100 – \$200	Mining, rock crushing, heavy trucking.
Direct Air Capture (DAC)	\$400 – \$800+	\$100 – \$250	Extreme electricity/heat use, custom chemical sorbents.